Sequence-Input Models for Variant-Effect Prediction in TEM-1 β-Lactamase

A literature review for the sequence-only β-lactamase functionality-prediction project (v2)

Scope and inclusion rule

In the clinic, the only input available for a novel antibiotic-resistance gene is its DNA sequence, translated to an amino-acid sequence. No crystal structure exists for a gene nobody has studied, and building a reliable multiple-sequence alignment (MSA) for a fast-evolving resistance determinant is neither guaranteed nor quick. The deployable question is therefore narrow and honest: *how well can a model predict whether a variant of a resistance enzyme is still functional, using the amino-acid sequence as its only input?* This review surveys every model that meets one criterion — **it takes an amino-acid sequence as input**, and asks, for each, what biology it encodes, how it reasons about the effect of a substitution, and whether it emits features that a downstream classifier can consume directly from sequence.

The inclusion rule is about *input modality, not output*. A model that folds a sequence into a 3D structure is still sequence-input and is analyzed here; a model that requires a solved or predicted structure, or an MSA, as its input is not deployable from sequence alone and is excluded (with the reason stated). This distinction is the spine of the whole project: it separates what a clinical tool could actually run from what merely performs well in a benchmark that assumes inputs the clinic does not have.

The central thesis this review builds toward is the project's scientific justification: **protein language models encode biology — evolutionary and biophysical constraint learned across millions of sequences, whereas a classifier trained on residue identity alone learns position-specific patterns that do not transfer to protein regions it has never seen.** If that thesis holds, a language-model-derived feature should beat an identity-only baseline precisely where it matters most: on held-out regions of the protein, the situation a novel resistance gene presents.

What breaks TEM-1

TEM-1 is a class A serine β-lactamase that hydrolyzes the β-lactam ring of penicillins, conferring resistance in *E. coli* and many other Gram-negative pathogens; the standard residue-numbering convention for class A β-lactamases is the Ambler scheme, which this project uses for reporting (Ambler 1991). A missense substitution can abolish resistance through two broadly distinct routes, and keeping them separate matters for interpreting every model below. The first is **loss of fold stability**: a substitution in the hydrophobic core or at a structurally critical position destabilizes the protein, it fails to fold or is degraded, and no functional enzyme reaches the periplasm. The second is **loss of catalytic function with**

the fold intact: a substitution at or near the active site (the S70 nucleophile, the K73/S130/E166 catalytic machinery) leaves a folded protein that no longer turns over substrate. A sequence-only model that scores a variant as "damaging" may be responding to either signal, and a recurring theme of this review is which models conflate them and which can, in principle, distinguish them.

The ground truth for training comes from deep mutational scanning (DMS), which measures the fitness of thousands of variants in parallel, and TEM-1 is among the most thoroughly scanned enzymes in biology. Jacquier and colleagues first captured its mutational landscape by systematic mutagenesis and phenotypic selection, establishing that most single substitutions are tolerated while a structured minority abolish function (Jacquier 2013). The primary training resource for this project is the Firnberg (2014) comprehensive TEM-1 fitness map, which extended that landscape to nearly all single amino-acid substitutions across the gene under ampicillin selection and produced a high-resolution, MIC-linked genotype–fitness landscape (Firnberg 2014). A second TEM-1 DMS study by Stiffler and colleagues added a consequential observation: measured fitness depends strongly on selection pressure, with purifying selection sharpening at higher antibiotic concentration, so the same variant reads as functional or dead depending on the ampicillin dose (Stiffler 2015). That dose-dependence is not a nuisance; it means any single hard threshold turning continuous DMS fitness into a binary functional/non-functional label is a modeling choice, not a given, and this project keeps that choice explicit and its label schemes swappable. These experiments are the reason a supervised sequence-only model is even possible here: they supply the functional labels that a language model, trained without any TEM-1 labels, is then asked to recover.

A second warning from the β-lactamase literature is that fitness is not a smooth, additive function of sequence. Epistasis pervades protein evolution (Starr & Thornton 2016), and much of the apparent nonlinearity in TEM-1-type landscapes is "global epistasis," a saturating nonlinear map from an additive latent trait to observed fitness (Otwinowski 2018). For this project the practical consequence is that a model predicting observed functionality is partly predicting a saturating nonlinearity rather than pure per-residue tolerance, and that any future extension to double and triple substitutions will not be a simple sum of single-mutant effects.

How sequence models learn constraint

The idea that mutation effects are readable from evolutionary sequence statistics predates language models. EVmutation showed that pairwise epistatic models fit to a multiple-sequence alignment predict mutation effects markedly better than site-independent conservation, because they capture coupled positions (Hopf 2017); DeepSequence replaced the pairwise model with a deep latent-variable generative model of the alignment and improved further (Riesselman 2018). The through-line is that a model of what natural sequences look like is implicitly a model of which mutations are tolerable, exactly the assumption behind using surprisal as a disruption proxy. Language models generalized this from one alignment to all of protein space. The foundational observation is that a neural network trained only to predict masked or next amino acids across a large protein database internalizes structure and function as a byproduct of learning the statistics of sequences. Rives and colleagues showed that scaling an unsupervised transformer on 250 million sequences causes representations of secondary structure, tertiary contacts, and functional sites to emerge in the embedding space without any structural supervision (Rives 2021). The mechanism is evolutionary: positions that are conserved across the training distribution are ones where substitutions were historically purged by selection, so the model learns to assign them low probability of change. When such a position is mutated, the model is "surprised," and that surprisal is a proxy for how disruptive the substitution is likely to be — which is exactly the quantity a variant-effect predictor needs.

Two things follow. First, a model's per-position probability distribution over the twenty amino acids is a direct readout of learned constraint, and the change in log-probability between the wild-type and mutant residue — the *masked-marginal score*, is a zero-shot variant-effect prediction requiring no labels at all. Meier and colleagues formalized this and showed that language-model log-likelihood ratios predict mutational effects across many DMS datasets, establishing the masked-marginal approach as the standard sequence-only scoring method (Meier 2021). Second, because the constraint the model learns is drawn from evolution, its signal is strongest for the *stability and generic-function* component of fitness and weakest for idiosyncratic, protein-specific catalytic requirements that evolution has not repeatedly tested across homologs, a limitation returned to below.

Sequence-input models analyzed

The ESM family (all included, ESMFold included)

The ESM family is the core of this project's candidate set, and the entire family qualifies under the input-modality rule — including ESMFold, because ESMFold takes a sequence as input and uses an ESM-2 language-model trunk to produce a structure.

ESM-1b (Rives 2021) is the original evolutionary-scale transformer and the proof that biological structure emerges from unsupervised sequence modeling. **ESM-1v** (Meier 2021) is an ensemble specifically trained and validated for zero-shot variant-effect prediction; it is the reference implementation of masked-marginal scoring and a natural single-model baseline for the resistance task. **ESM-2** (Lin 2023) is a larger, better-trained successor released in sizes from 8M to 15B parameters (the project's default checkpoint, `esm2_t33_650M_UR50D`, is the 650M variant); its representations improved enough that a folding head could be attached directly. That folding head is **ESMFold**, described in the same work: it predicts atomic-level structure from a single sequence with no MSA, at a fraction of AlphaFold2's runtime, by reading structure out of the ESM-2 trunk (Lin 2023). This is why ESMFold belongs in the sequence-input set and why ESM-2 remains a feature source even though ESMFold is a structure predictor: the same trunk that ESMFold folds *with* yields, on its own, the per-position logits and embeddings that this project consumes as features. **ESM C (ESM Cambrian)** is EvolutionaryScale's more recent, compute-efficient representation-focused family, released as a versioned model rather than a peer-reviewed paper (EvolutionaryScale 2024); it is included as a candidate because it reports strong representation quality at lower cost, though its non-standard release means performance claims should be treated as vendor-reported until independently benchmarked. **ESM3** (Hayes 2025) is a multimodal generative model over sequence, structure, and function tokens; it is sequence-input-capable and noted for completeness, though its generative multimodal design is heavier than this project needs for scoring.

For the resistance task, the practical readouts from the ESM models are three: the masked-marginal scalar per variant (zero-shot effect score), the per-position probability vector across all twenty amino acids (a richer constraint fingerprint), and the mean-pooled or per-residue embedding (a learned representation a classifier can be trained on). All three are computed directly from the TEM-1 sequence.

ProtT5, ProstT5, and VESPA

The ProtTrans effort trained several transformer architectures on protein sequences, of which **ProtT5** (an encoder-decoder T5 trained on UniRef) produces the most widely used embeddings and is competitive with the ESM models on many downstream tasks; critically, its per-residue embeddings capture conservation and structural context from sequence alone (Elnaggar 2022). Built on ProtT5, **VESPA** is a variant-effect predictor that derives a substitution score from language-model embeddings plus a learned conservation prediction, explicitly targeting the missense-effect problem this project addresses (Marquet 2021). VESPA is a strong sequence-only comparator because it is purpose-built for variant effects rather than repurposed from a general LM. **ProstT5** extends ProtT5 into a bilingual sequence↔structure model that translates between residues and Foldseek 3Di structure tokens (Heinzinger 2024); it is sequence-input (it can take sequence and emit structure tokens) and is therefore analyzed here, but the moment its *structure-token output* is fed back as a feature it becomes a structure-derived pipeline, so for this project ProstT5 is relevant as a sequence embedder, not as a 3Di-feature source.

Autoregressive generative models: ProGen, ProGen2, ProtGPT2

A second family scores variants by generative likelihood rather than masked reconstruction. **ProGen** demonstrated that a large autoregressive LM conditioned on functional tags generates functional enzymes, and its sequence likelihoods correlate with fitness (Madani 2023). **ProGen2** scaled this and studied the fitness-prediction and design boundaries of autoregressive protein LMs, reporting competitive zero-shot variant-effect performance (Nijkamp 2023). **ProtGPT2** is a GPT-2-style unsupervised model whose primary use is de novo design but whose perplexity is likewise a sequence-only fitness proxy (Ferruz 2022). These are all sequence-input and feature-eligible (via per-sequence or per-position likelihood), and are included for breadth, though for a small single-domain enzyme the masked-LM ESM models are generally the stronger and cheaper variant-effect scorers.

Efficient and alternative architectures: CARP, Ankh, UniRep, ProteinBERT

CARP shows that convolutional architectures are competitive with transformers for protein-sequence pretraining, offering a cheaper feature extractor whose per-position representations rival transformer embeddings on several tasks (Yang 2024) — relevant here because the project runs on CPU and inference cost is a real constraint. **Ankh** is a compute-optimized protein LM that reaches strong general-purpose performance with fewer parameters through architectural and training-objective choices (Elnaggar 2023). **UniRep** is the early mLSTM-based representation learner that first showed a single recurrent embedding could support rational protein engineering (Alley 2019), and **ProteinBERT** is a compact BERT-style model jointly trained on sequence and Gene Ontology annotations that produces useful whole-sequence and per-residue features cheaply (Brandes 2022). These four are all sequence-input, feature-eligible, and worth including as lighter-weight alternatives when the 650M–15B ESM checkpoints are too expensive.

Sequence-input structure predictors: AlphaFold2, AlphaFold3, OmegaFold

These fold a sequence into coordinates and are analyzed because they are sequence-input, but a caveat governs their use in *this* project. **AlphaFold2** achieves near-experimental accuracy but does so by consuming an MSA as part of its input pipeline (Jumper 2021); in the strict sense its accuracy depends on alignment input, which is exactly the resource the clinic may lack, so while it is reviewed as a landmark it is not a strictly single-sequence model. **AlphaFold3** generalizes to protein, nucleic-acid, and ligand complexes with a diffusion-based architecture (Abramson 2024). **OmegaFold** is the closest to a true single-sequence folder, predicting high-resolution structure from one sequence without an MSA by pairing a protein LM with a geometry module (Wu 2022). For this project these models are reviewed as sequence-input, but their *output is coordinates, not features* — turning a predicted structure into a classifier input requires a structural-feature extraction step, which moves the pipeline out of the direct-from-sequence feature regime the benchmark is built on. ESMFold is the exception that stays in the feature set, because its ESM-2 trunk yields sequence features without ever touching the folded coordinates.

Excluded models and why

Three otherwise-excellent models are excluded because they require an input the clinic cannot assume. **SaProt** achieves strong variant-effect performance using a structure-aware vocabulary in which each token fuses an amino acid with a Foldseek 3Di structural token (Su 2024); those 3Di tokens are extracted from a 3D structure via Foldseek (van Kempen 2023), so SaProt cannot run on a bare sequence — it needs a structure (real or predicted) in the loop, making it a sequence-*via*-structure pipeline rather than a direct sequence→feature model. **MSA Transformer** conditions on a multiple-sequence alignment as its input and reasons over evolutionary covariation across the alignment columns (Rao 2021); no alignment, no prediction. **EVE** is a deep generative model trained per-protein on an MSA and is a leading clinical variant-effect predictor, but its dependence on a curated family alignment disqualifies it as sequence-only (Frazer 2021). **Tranception** is a hybrid whose base is a sequence-only autoregressive transformer, but its best-reported performance comes from inference-time *retrieval* that queries an MSA (Notin 2022); the retrieval mode is excluded as alignment-dependent, though the point that its non-retrieval base is sequence-only is worth keeping in mind as a fallback comparator. These exclusions are not judgments of quality, SaProt and EVE are among the most accurate variant-effect models published, but of *deployability under the sequence-only constraint*.

Stability signal versus catalytic signal

The most important biological caveat for interpreting every included model is that language-model surprisal predominantly captures the *stability and generic-conservation* component of a variant's effect, and only indirectly captures protein-specific *catalytic* requirements. This is not speculation: language-model and structure representations predict thermodynamic stability (ΔΔG) well (Blaabjerg 2023), and a large fraction of apparent loss-of-function in mutational data traces to loss of stability rather than direct active-site disruption (Cagiada 2021). The reason is mechanistic: a masked LM assigns low mutation probability to positions that are conserved across the training distribution, and the positions most consistently conserved across a protein family are the folding core and the broadly shared functional scaffold. A catalytic residue unique to one enzyme's chemistry — or a second-shell residue that tunes an active site in a way evolution has not repeatedly re-tested across homologs, may carry weaker LM constraint than its functional importance warrants. This predicts a specific failure mode for TEM-1: the models should score core-destabilizing substitutions well, and may under-weight active-site substitutions that abolish catalysis while leaving the fold intact. The project's design responds to this directly. Because the DMS labels reflect *total* functional loss (both routes), a supervised classifier trained on language-model features can learn to reweight the raw surprisal toward the catalytic positions that matter for TEM-1 specifically, which is the sense in which the classifier "learns the biology the language model teaches it," rather than trusting the zero-shot score alone. It is also why physicochemical features that flag charge and size changes at active-site positions are worth concatenating: they add a signal orthogonal to conservation.

The generalization argument

Building a supervised classifier on top of an unsupervised backbone, which is exactly this project's design, is well-precedented. UniRep first showed learned embeddings feed downstream property predictors (Alley 2019); Gelman and colleagues trained neural networks directly on DMS data and analyzed how sequence-function models learn (Gelman 2021); and ECNet showed that combining evolutionary context with local sequence features beats embeddings alone on supervised fitness prediction (Luo 2021). The recurring finding is that a supervised model on good representations beats a zero-shot score when tested on held-out mutations from the same protein, which raises the methodological question this project foregrounds: generalization across sequence *position*.

A variant-effect model can succeed for the wrong reason. If training and test variants are drawn at random from the same positions, a classifier can memorize which *positions* are intolerant and score any substitution there as damaging, achieving high accuracy without any transferable understanding of the substitution itself. This is position leakage, and it is exactly what an identity-only one-hot baseline exploits. The benchmark literature documents the effect directly: FLIP built fitness-landscape tasks around realistic splits that withhold whole positions or regions rather than random mutations, and showed performance drops sharply under these harder splits (Dallago 2021). That is the direct justification for this project's contiguous-split strategy. The three split schemes this project uses are designed to expose it: a **random split** scatters variants across positions and is the most leakage-prone (optimistic); a **modulo split** assigns positions by index modulo k, reducing but not eliminating adjacency leakage; and a **contiguous split** withholds entire stretches of the protein, so the model must score substitutions in regions whose positions it never saw in training. The contiguous split is the honest analog of the clinical situation — a novel resistance gene is an unseen region, and is therefore the project's headline metric. The thesis of the whole effort is testable precisely here: a language model that encodes transferable biology should retain accuracy on held-out regions, while an identity-only baseline, having only memorized positions, should collapse. The gap between random-split and contiguous-split accuracy is the "leakage tax," and making it visible is a deliberate reporting choice.

Where this sits in the variant-effect field

The broader field has converged on the finding that unsupervised sequence and evolutionary models predict variant effects well enough to matter clinically. DeepSequence established that deep generative models of evolutionary data capture mutational effects (Riesselman 2018); EVE extended this to disease-variant classification (Frazer 2021); and AlphaMissense combined a structure-informed language model with population data to produce proteome-wide pathogenicity scores now used in clinical interpretation (Cheng 2023). ESM-1b has been applied genome-wide to predict the effects of coding variants directly from sequence (Brandes 2023), and general protein LMs have guided real protein engineering, including antibody affinity maturation from sequence alone (Hie 2023). How such claims should be reported is itself a settled lesson: head-to-head DMS benchmarking repeatedly finds that apparent rankings shift once evaluation is standardized (Livesey 2020; Livesey 2023), so a single AUROC on a favorable split is not a result — performance across split strategies, with the hardest reported prominently, is. The standardized arena for comparing these methods is ProteinGym, a large benchmark of DMS and clinical assays with defined splits that this project's evaluation design deliberately mirrors (Notin 2023). What distinguishes the present project from most of this literature is its refusal to use the inputs the clinic lacks: no experimental structure, and, in the strict candidate set carried into the benchmark, no MSA. The clinical framing for why a *functional* rather than merely *present* resistance call matters is the same one that motivates sequence-variant interpretation standards generally (Richards 2015).

Bottom line

Every model that reads an amino-acid sequence as input has been analyzed, and they divide cleanly by what they can contribute to a sequence-only resistance-prediction tool. The ESM family — ESM-1v and ESM-2 in particular, with ESM C as an efficiency candidate and ESMFold's trunk as a legitimate feature source, provides masked-marginal surprisal, per-position constraint vectors, and embeddings directly from sequence, and is the strongest, best-validated candidate set. ProtT5/VESPA offer a purpose-built variant-effect comparator; CARP, Ankh, ProteinBERT, and UniRep offer cheaper feature extractors that matter on CPU; the autoregressive models add likelihood-based scores for breadth. The sequence-input structure predictors (AlphaFold2/3, OmegaFold) are reviewed as landmarks but are not direct feature sources, and SaProt, MSA Transformer, EVE, and retrieval-Tranception are excluded for requiring structure or alignment inputs the clinic cannot assume. The single most important interpretive caveat carried into modeling is that language-model surprisal captures stability and generic conservation more faithfully than enzyme-specific catalysis, which is the gap a supervised classifier trained on DMS labels, plus orthogonal physicochemical features, is meant to close. And the honest measure of success is not random-split accuracy but performance on held-out contiguous regions, the only split that tests whether a model has learned transferable biology rather than memorized which positions of TEM-1 are fragile.

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